

In general, a statement involving the n variables $x_1, x_2, \dots, x_n$ can be denoted by $P(x_1, x_2, \dots, x_n)$.

A statement of the form $P(x_1, x_2, \dots, x_n)$ is the value of the propositional function P at the n -tuple $(x_1, x_2, \dots, x_n)$ and P is also called an n -place predicate or an n -ary predicate.

Preconditions and Postconditions

Predicates are also used to establish the correctness of computer programs, that is, to show that computer programs always produce the desired output when given valid input. (Note that unless the correctness of a computer program is established, no amount of testing can show that it produces the desired output for all input values, unless every input value is tested.)

The statements that describe valid input are known as preconditions and the conditions that the output should satisfy when the program has run are known as postconditions.

Example 7: Consider the following program, designed to interchange the values of two variables x and y .

temp := x

x := y

y := temp

Find predicates that we can use as the precondition and the postcondition to verify the correctness of this program. Then explain how to use them to verify that for all valid input the program does what is intended.

Soln: For the precondition, we need to express that x and y have particular values before we run the program. So, for this precondition we can use the predicate $P(x, y)$, where $P(x, y)$ is the statement " $x = a$ and $y = b$ ", where a and b are the values of x and y for all input values, for the postcondition we can use $Q(x, y)$, where $Q(x, y)$ is the statement " $x = b$ and $y = a$ ".